PERFORMANCE IMPROVEMENT OF HORIZONTAL AXIS WIND TURBINE USING ACTIVE CONTROL METHODS

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Abstract

The objective of the paper is to improve the performance of horizontal axis wind turbine using the various active control methods. The numerical analysis of injection, suction method was carried out with the Reynolds average Navier stokes equation using the k epsilon turbulence model. The results show that the suction method works more efficiently than injection and the baseline aerofoil. It disclosed the stall improvement of 100% and suction lift coefficient of 100% and injection lift in improvement up to 60%. The novelty of the proposed work is the HAWT wind turbine blade analysed separately using the injection and suction method, and then the performances are compared. The proposed study can be used for future model of horizontal and vertical axis wind turbine to enhance the performances at different angles.

1 Introduction

Nowadays, the majority of the countries use wind turbines for power generation to meet future demand. There are two different types of wind turbines like horizontal and vertical axis wind turbines. In that horizontal axis wind turbine used in most places to enable effective power generation. In worldwide the majority of the researcher working on HAWT to improve the performances using the various technique's. In that one of the effective method to enhance the aerodynamic performances using the active method of boundary layer control. The upcoming literature shows that the various boundary layer control methods used for HAWT.

Sun, Y.,et al [1] used single-channel jet control strategy with suction and injection slots effectively suppresses reversed flow and improves net power output without requiring additional air. Numerical simulations confirm that this strategy enhances power output by up to 45.68% at low wind speeds, especially at low tip speed ratios. Wang, P., et al [2] effectively suppresses flow separation and improves aerodynamic performance by enhancing pressure difference and stability on a wind turbine. Numerical simulations show that BLS control boosts power extraction, particularly at low to medium tip speed ratios, when applied to a horizontal-axis wind turbine. Akhter, M. Z., & Omar, F. K [3] growth drives demand for larger turbines, but increased blade fatigue and maintenance costs raise the cost of energy. Flow-control devices, through passive and active control mechanisms, can improve aerodynamic performance, reduce loads, and help lower operational costs, offering a potential solution to these challenges. Chawla, J. S. [4] investigates the use of surface suction-based active flow control to improve aerodynamic efficiency and reduce structural loads in small wind turbines operating at low Reynolds numbers. The approach aims to prevent flow separation and lower the Cost-of-Energy, particularly in off-grid turbines operating in low wind speed environments. Moshfeghi, M., et al[5] investigates the effects of a passive flow control method on horizontal axis wind turbines by splitting the blades and analysing the impact on power coefficient, torque, and flow patterns. Results show that split location significantly influences torque generation, especially under different flow conditions and tip speed ratios, with the split blade contributing more power at lower tip speed ratios. Ding, N., et al [6] presents an active flow control technique using a combination of plasma jet actuators (PJA) and synthetic jet actuators (SJA) to enhance the aerodynamic lift of fixed-wing UAVs. Results show that the combined PAS control mode significantly increases lift, with the main wing contributing most to the lift enhancement, and flow control parameters such as plasma voltage and synthetic jet exit velocity influencing the lift performance. McBreen, B., et al [7] study investigates the effects of high enthalpy flow in Co-Flow Jet (CFJ) active flow control using IDDES, showing that the "Hot Jet" CFJ increases the lift coefficient by 4.8% compared to the "Cold Jet" but requires higher power and leads to a slight decrease in aerodynamic efficiency. The enhanced turbulence from the high enthalpy jet improves mixing and vortex formation, boosting aerodynamic performance but at the cost of higher drag and power demand. Ibrahim, W. M. [8] discusses the importance

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of controlling rotor blade flow separation and dynamic stall to improve horizontal axis wind turbine (HAWT) performance, comparing active and passive flow control methods. While active control requires additional power and equipment, passive flow control is simpler, more cost-effective, and beneficial in real-world applications, with CFD playing a key role in optimizing turbine designs. Vijayan, S. N., et al [9] develops a composite material using LM13 aluminium alloy and zirconium diboride (ZrB2) to enhance mechanical properties, with applications in automotive industries. Similar composite materials can be adapted for horizontal axis wind turbines to improve structural integrity and performance under varying mechanical stresses. He, P., & Xia, J. [10] integrates the control system into blade element momentum theory (BEMT) and fluid-structure interaction (FSI) calculations to evaluate its impact on aerodynamic characteristics and elastic deformations of wind turbines. The control system, including variable speed and pitch control, significantly influences turbine performance, especially in varying wind conditions and during larger blade deformations. Gore, M. R., & Khandal, S. [11] analyses the impact of vortex generators (VGs) on symmetrical and cambered aerofoils, showing that VGs improve stall AOA by 4° (simulation) and 2° (experiment) without increasing drag. The study demonstrates enhanced aerodynamic efficiency for wind turbines. Saranya, S., et al [12] investigates active and passive flow control mechanisms to improve aerodynamic performance, showing that active methods like Improved Blowing Suction System (IBSS) significantly enhance efficiency. Similar techniques can be applied to horizontal axis wind turbines to optimize aerodynamic performance and control the boundary layer. Yang, Y., et al [13] investigates the use of vortex generators to improve the aerodynamic performance of wind turbine blades under rough wall conditions, showing that they delay flow separation and increase lift-to-drag ratios. The results demonstrate a 47.8% increase in power coefficient by reducing flow separation, highlighting the potential of vortex generators to counteract surface roughening effects and enhance wind turbine efficiency.

Zha, R., et al [14] reviews the performance calculation and design methodologies for horizontal-axis wind turbine blades, focusing on aerodynamic, structural, and aero-structural design. It emphasizes the importance of selecting the most suitable solution from a range of options and highlights the need for future research on ultra-long, flexible blades, multi-objective optimization, and the challenges of deep-sea wind turbine development. Balaji, K., et al [15] research identifies the optimal quarter-chord injection site for improving aerodynamic efficiency, resulting in an 8% lift increase and a 43.5% drag reduction. Similarly, this approach could be applied to horizontal axis wind turbines to enhance performance and reduce drag, improving energy capture efficiency. Balaji, K., & Wessley, G. [16] conducts a numerical analysis of the NACA 6321 aerofoil using three turbulence models, with the k-omega model showing the best correlation with experimental data, especially at low and high angles of attack. The k-omega model accurately predicts turbulence properties with less than 4% error compared to experimental findings. Saini, H., et al [17] explores the use of Gurney flaps on small horizontal axis wind turbines to enhance aerodynamic performance, showing increased power coefficients at higher tip speed ratios. While the Gurney flaps improve performance, they also introduce a trade-off with increased structural stress and noise, demonstrating their potential as a passive flow control method. Yang, C., et al [18] investigates the dynamic aerodynamic performance of a 3.3 MW wind turbine blade during variable pitch using numerical simulations, showing that the optimal pitch angle increases with wind speed. The results also reveal that pitch changes impact torque and thrust coefficients, while enhancing the average wind speed at the turbine wake by 10.7% in constant-wind-speed scenarios. Hamlaoui, M. N., et al [19] introduces an advanced design technique for determining radial local chord distribution on Horizontal Axis Wind Turbine (HAWT) blades using an inverse CFD actuator disk method, resulting in enhanced aerodynamic performance. The optimized rotor design improves power coefficient by 70%, annual energy production by 17.64%, and reduces velocity deficit by 15%, offering significant benefits for wind turbine efficiency and wind farm optimization. Tanovic, D., et al [20] investigates the effect of blade rotation angle on the power coefficient, torque, and mechanical power in small horizontal axis wind turbines (HAWT), using both numerical analysis and experimental methods. The study shows that increasing the rotation angle improves performance within a specific range of the tip speed ratio and highlights the influence of rotor and generator design on torque and mechanical power characteristics.

The above literature shows the various active and passive methods used for HAWT. It shows that the active methods majority used the combination of injection and suction mechanism and parameters study of injection and suction locations. The passive methods used the Vortex generators, dimples. There is no evidence of papers studied alone the injection or suction methods for this study. Based on this, the objective of the paper is to use the injection and suction methods alone with the location of the mid chord.

2. Methodology

The numerical study was carried out with injection and suction aerofoils, and the results were compared with the baseline aerofoils. The analysis was carried out using the pre-processing to post-processing procedure.

2.1 Design

The simulation study uses the aerofoil identified from the market availability as well as the well-known S809 aerofoil with the dimension of 600mm as shown in Fig 1a. The injection model places the injection point at the mid of the chord with the height of 12mm as shown in fig 1b.In a similar manner, the suction is placed with the same mid-section of the aerofoil with the same height similar to injections as shown in fig 1c. The domain used for this study is shown in Fig. 2.

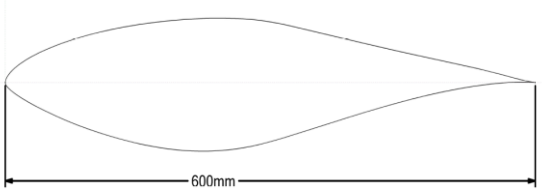

Fig 1a: Baseline aerofoil

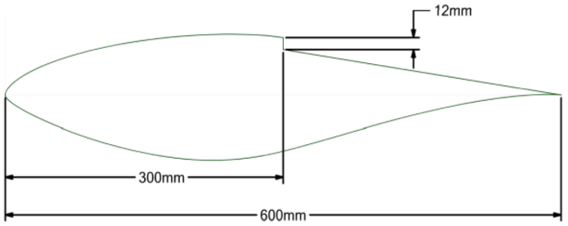

Fig 1b: Injection Model

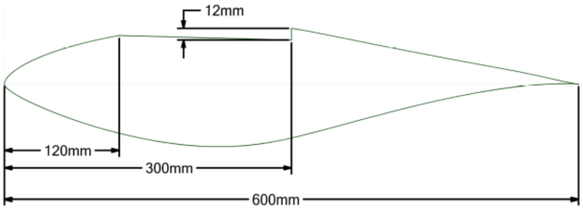

Fig 1 c: Suction aerofoil

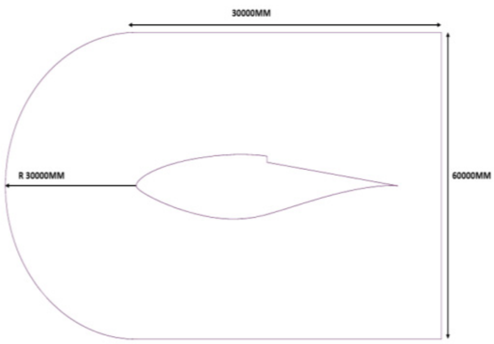

Fig 2: Domain used for this study

2.2 Meshing

The designed model, along with the domain, is converted into small elements and nodes. The unstructured mesh with the quadrilateral elements is used for this study. The amount of nodes and elements, as shown in fig 3, are finalized from the grid independent study mentioned in the literature [2].In the meshing, the entire domain is named into the components, which can be used for boundary conditions.

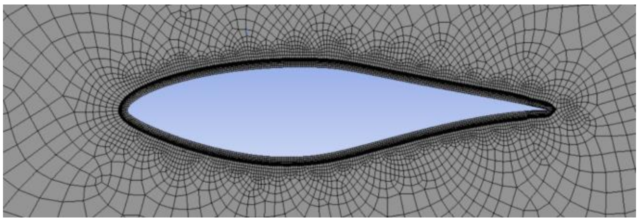

Fig 3: Close view of mesh

2.3 Validity

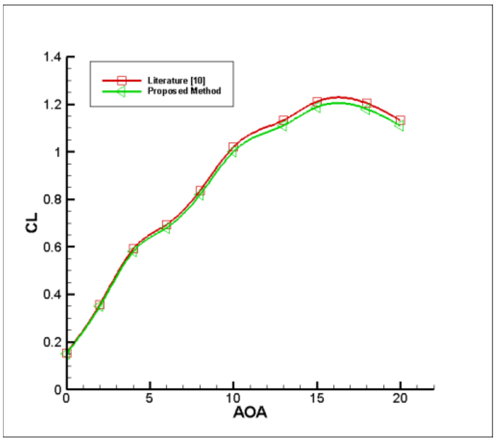

Fig 4: Validation Graph

Fig 4 shows the validation graph of the proposed model with the existing literature [10]. It indicates that both models are following the same track with an error percentage of 4%. It concluded that the proposed method followed the equation and that boundary conditions can be used for further variation of the study. Both the lift and drag coefficient curves followed the same trend, and this provided the confidence to take the procedure forward. Similar results were observed in the literature [16].

2.3 Numerical Analysis:

The two-dimensional analysis of the designed aerofoil used the Reynolds average Navier Stokes equation with k epsilon turbulence model. The pressure-based solver with the convergence value of 10-6 is chosen for the study. The boundary conditions used for the injection and suction aerofoil are described in figures 5a & b. The analyses are carried out with real-time conditions with different angles of attack.

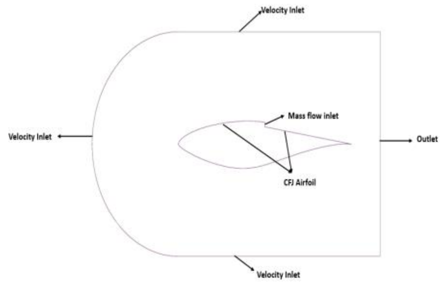

Fig 5a: Injection Boundary Conditions

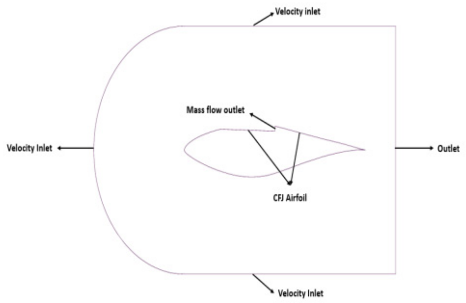

Fig 5b: Injection Boundary Conditions

3 Results and Discussion

The baseline, injection and suction models are analysed using the computational fluid dynamics software with different operating conditions, which are analysed and explained in the upcoming portions. The results are discussed in graphical and visualization manners.

3.1 Lift Coefficient

Fig 6 shows the analysis of aerofoil behaviour with various angles of attack of baseline, injection, and suction aerofoils. It describes that the lift increases based on the angle of attack up to the stall later, and the lift decreases. It indicates that the suction aerofoil is creating efficient performance than the injection and baseline aerofoil. Both active methods enhance the stall upto 100% at the same time the suction improve the lift improvement up to 219%, and the injection lift coefficient increases up to 60% that is similar to an aircraft wing [12, 15]. The injection introduces the secondary flow at the downstream to minimize the wake region and suction design effectively doing the suction process which removes the wake effectively and make flow more attached. This analysis proved that the active method effectively controls the boundary layer, as the suction method is more effective than the injections.

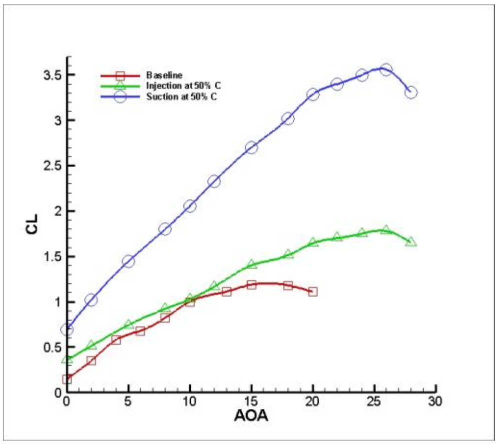

Fig 6: Co- efficient of lift curve

3.2 Drag Coefficient Curve

The drag coefficient values are predicted for various angles of attack, as shown in Fig 7. It discloses that the active methods are enhancing the drag coefficient while improving the angle of attack. The injection and suction methods are improving the drag by more than 100% compared to the baseline aerofoil. The reason behind that is lift-induced drag is created over the aerofoil in improving the lift, which concurrently increases the drag. The injection creates more drag compared to the suction method due to the less impact on the boundary layer effects. The entire analysis proved that the drag coefficient of active methods increases the more lift-induced drag.

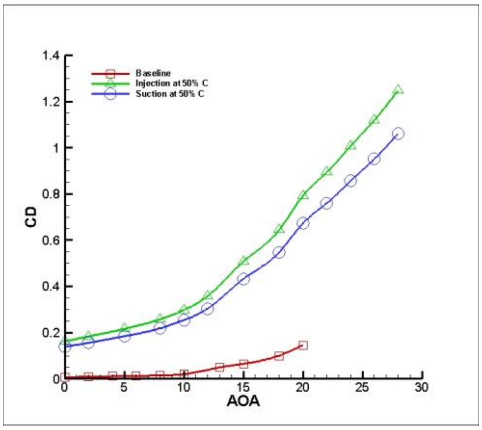

Fig 7: Co- efficient of drag curve

3.3 Visualization Curve

Fig 8 indicates the velocity and pressure contour of the baseline aerofoil and proposed methods. It indicates that in velocity contour clearly indicate that baseline aerofoil at downstream more wake region with the effect of boundary layer. At the same time, the suction and injection velocity profile reveals that the wake regions are minimized and effectively increased the lift coefficient values. The pressure contour clearly defined that the proposed active methods are creating the less pressure over the top of aerofoil compare to the Baseline aerofoil. Similar results were obtained in the literature [4, 5]

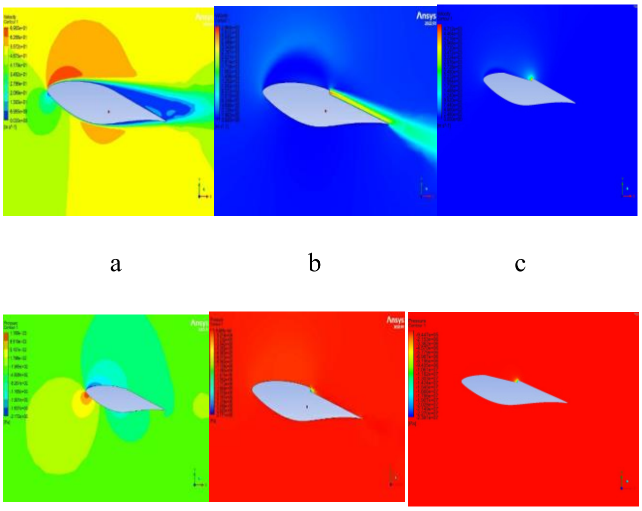

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Fig 8: Visualization Study

4 Conclusion

The S 809 aerofoil with the design of injection and suction models is designed and numerically investigated with various angles of attack. The obtained results are analysed using the various parameters through the graphical and visual methods studied at actual working conditions. It brought the following conclusions

- The proposed active and passive methods are used to improve the aerodynamic performances with the incremental of drag. It proved that the proposed models are effectively worked at various operating conditions

- Both the injection and suction methods enhance the stall angle up to 100% due to the effective control of the boundary layer from a lower to a higher angle. Especially at higher angles, this method works effectively

- The suction methods increase the peak lift coefficient by 100%, and the injection method enhances up to 60% compared to the baseline aerofoil. It concluded that the proposed active methods made the flow more attached and improved the performances

- In both methods, the drag values are increased more than 100% in comparison with the baseline aerofoil. The effect indicates that the proposed methods are creating the lift-induced drag over an aerofoil.

The overall study of the research proved that the active methods can effectively work over a wind turbine. In the future, the same model can be analysed with different mass flow and angles and can provide a detailed way to increase the performance with minimum drag increment.

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